Wildfire Simulation & AI Agent Evaluation

Technical Project Report | Texas State University CADS Workshop | October 2026

Portfolio project based on “How to Build Agents for Research,” workshop materials by Aniruddha Bora. Prepared from the supplied workshop code, assignment notebook, and independently recorded simulation results.

1. Executive Summary

This project investigates how wind, fuel dryness, and river-crossing closures affect simulated wildfire damage to a town. Using a probabilistic grid-based wildfire simulator and Python analysis, 100 seeded fire realizations were evaluated per scenario. Under the default scenario, an average of 31.56 of 36 town structures were reached. Closing the middle and south crossings together reduced the simulated average to 1.9 structures, a reduction of approximately 94.0%. The project also examines how AI-agent recommendations should be tested independently, rather than accepted solely because they appear convincing.

2. Objectives

- Run a reproducible wildfire baseline and characterize uncertainty.
- Compare wind directions, wind speeds, dryness conditions, and crossing-closure strategies.
- Identify the best intervention among options constrained to closing at most two of three crossings.
- Evaluate agent trace quality and document which agent-based experiments remain unexecuted.

3. Tools and Model

Python, NumPy, pandas, Jupyter/Google Colab, fire_sim.py, and agent_tools.py. The workshop simulator models a 48 × 72 grid with 100 m × 100 m cells (1 hectare each), a river with north, middle, and south crossings, and a town represented by 36 structures. Fire propagation depends on fuel, slope, wind, and dryness; ember spotting is also modeled. The scenario is illustrative and is not calibrated for real-world emergency planning.

4. Experimental Method

The supplied simulator was used with 100 Monte Carlo runs per option and a fixed seed of 0 for controlled comparisons. Baseline conditions were wind speed 6 m/s toward the east, dryness 0.9, and a 55-step horizon. Each scenario was summarized by mean structures reached; selected tests also tracked burned area and town-reach probability. Closure combinations were enumerated under the two-crossing crew limit. A supplied flawed agent trace was critiqued against the independent numerical checks.

5. Results

Baseline

Average structures lost: 31.56 / 36. Town reached in 96 of 100 runs (Wilson 95% confidence interval approximately 0.90–0.98). Mean area burned: 2,536.7 hectares.

Wind Direction

Eastward wind produced the highest average structure loss (31.6). Southeast produced 21.9, northeast 8.8, and the other tested compass directions produced 0.0 under these specific model settings.

Wind Speed

At 2, 4, 6, 8, 10, and 12 m/s, mean structures lost were respectively 31.3, 31.8, 31.6, 32.3, 32.6, and 33.5. Mean area burned declined across the same sequence from approximately 2,655 to 1,846 hectares. This counterintuitive outcome illustrates why different performance measures must be reported separately; it is not a general claim about real wildfires.

Crossing Intervention Comparison

Crossings closed	Mean structures reached (of 36)
None	31.6
North	31.4
Middle	12.6
South	29.3
North + middle	12.2
North + south	29.4
Middle + south	1.9

Best tested strategy: middle + south. Relative reduction compared with no closure: (31.56 − 1.9) / 31.56 ≈ 94.0%. These values are sample averages, not guaranteed outcomes.

Dryness Sensitivity

Closing the middle crossing alone reduced mean loss by at least 50% at tested dryness values 0.6–0.9, but not at 1.0–1.3. Thus 0.9 is the highest tested dryness meeting that criterion; it is not a precisely estimated physical threshold.

6. Agent Verification and Limitations

The supplied example of a flawed agent trace tested only four of seven permitted closure options, used ten rather than 100 runs per option, and varied seeds between alternatives. Its conclusion was therefore inadequately supported. The prepared notebook contains agent prompts, but live agent responses, Exercise 3 trace.json, and the ten-run agent reliability evaluation were not executed or verified. They should not be claimed as completed. The example agent's text-based code guard is not a secure sandbox.

7. Conclusions

Within the workshop's simulated environment, closing the middle and south crossings was the most effective tested intervention. The study demonstrates scenario design, repeated simulation, controlled comparisons, interpretation of uncertainty, and the importance of independently verifying AI-generated recommendations. Future work should execute the agent in an isolated environment, save traces, compare its answers to the numerical benchmark, and test robustness over additional seeds and parameter ranges.

8. GitHub / Résumé Description

Project title: Wildfire Simulation and AI Agent Evaluation (Python, NumPy, Jupyter/Colab)

- Analyzed a 48 × 72 probabilistic wildfire model using reproducible 100-run scenario comparisons across wind, dryness, and intervention settings.
- Evaluated seven river-crossing closure strategies; identified a two-crossing plan that reduced average simulated structure loss from 31.6 to 1.9 (≈94%) under baseline conditions.
- Documented model uncertainty and critiqued an AI-agent trace for incomplete option coverage, inadequate sample size, and uncontrolled random seeds.

9. Attribution and Reproducibility

Source: Aniruddha Bora, "How to Build Agents for Research," CADS workshop, Texas State University, October 2026. Workshop-provided fire_sim.py, agent_tools.py, and assignment materials were used as the project foundation. See README.md, RESULTS.md, and wildfire_assignment.ipynb in the repository. Independent results were reported for n=100, seed=0; see notebook for implementation.